

ER Physiology Practice Questions with answers

Topic: L9-Calcium Homeostasis

1. A 61-year-old man comes to the physician because of a 6-week history of bone pain, dyspepsia, and a 3-kg (6.6-lb) weight loss. His vital signs are within normal limits. Physical examination shows dehydration and pallor. Laboratory studies show elevated serum calcium concentration, elevated urinary calcium concentration, and elevated serum PTH concentration. Radiography shows lytic lesions in the long bones and pelvis. Neck imaging studies show nodular lesion posterior to the right lobe of the thyroid gland. Histological examination of a biopsy specimen from this nodule shows increased mass of the chief cells. Hypercalcemia of this patient is best explained by the action of PTH through PTH receptors located in which of the following organs?

- A. Small intestine
- B. Bones
- C. Skeletal muscles
- D. Liver
- E. Parathyroid gland

2. A 49-year-old man comes to the physician for a follow-up examination. He has an 8-month history of end stage kidney disease who is waiting for a renal transplant. He is having hemodialysis twice a week. He complains of recurrent muscle cramps, bone pains, and fatigue. Laboratory studies show a serum creatinine of 6.1 mg/dL (N: 0.6-1.2), blood urea nitrogen of 76 mg/dL (15-25) and serum calcium of 6.2 mg/dL (N: 8.5-10.5). His glomerular filtration rate is 5 mL/min and has a significant loss of renal function. Which of the following best explains the altered serum calcium in this patient?

- A. Reduced synthesis of cholecalciferol in the skin
- B. Low serum PTH
- C. Low vitamin D intake
- D. Decreased serum albumin
- E. Reduced 1-alpha hydroxylase activity
- F. Reduced 25-hydroxylase activity

3. A 21-year-old man comes to the physician for a routine examination. He says that his elder brother is diagnosed with kidney stones because of a high serum calcium concentration. His vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show a serum calcium of 10.8 mg/dL (Normal: 8.5-10.5) and an elevated serum PTH concentration. A 24-hour urinary calcium excretion is lower than normal. He is diagnosed with a defect in calcium sensing receptors. These receptors are present on which of the following cells?

- A. Oxyphil cells of the parathyroid gland

- B. Chief cells of the parathyroid gland
- C. Parafollicular cells of the thyroid gland
- D. Osteoblasts of bone
- E. Osteoclasts of bone
- F. Small intestinal epithelial cells

4. A 31-year-old woman comes to the physician because of a 4-week history of recurrent episodes of muscle cramps and sensory disturbance in her hands and legs. She has a 2-year history of malabsorption. She recently moved from tropical to temperate region where she is least exposed to sunlight. Physical examination shows no abnormalities. Laboratory studies show a serum calcium concentration of 8 mg/dL (Normal: 8.5-10.5) and a low serum concentration of vitamin D. She develops muscle cramps when physician asks her to perform hyperventilation for a few minutes. Which of the following mechanisms best explains this patient's presentation following hyperventilation?

- A. Increased PaO₂ reduces ionized serum calcium
- B. Decreased PaO₂ increases ionized serum calcium
- C. Increased PaCO₂ reduces ionized serum calcium
- D. Decreased PaCO₂ reduces ionized serum calcium
- E. Decreased PaCO₂ increases ionized serum calcium

5. A 26-year-old woman comes to the physician because of an 8-month history of recurrent episodes of muscle cramps, non-bloody diarrhea, and generalized bone pains. She says that diarrhea is accompanied with right lower quadrant pain. Physical examination shows tenderness in the long bones of her upper and lower extremities. Laboratory studies show a serum calcium concentration of 6.8 mg/dL (Normal: 8.5-10.5) and a low vitamin D concentration. Histological examination of a small-bowel biopsy specimen shows reduced absorptive surface in the small intestinal epithelium. She is diagnosed with Osteomalacia due to reduced absorption of vitamin D in the gastrointestinal tract. Which of the following sets of additional laboratory alterations is most likely in this patient?

Option	Serum PTH	Serum Phosphate	Serum Glucose
A	↑	↑	↑
B	↓	↓	↓
C	↑	↓	↑
D	↓	↑	Normal
E	↑	↓	Normal

6. An 8-year-old girl is brought to the physician by her mother because of a 6-month history of recurrent episodes of muscle cramps. She has a family history of similar

symptoms associated with skeletal deformities. Physical examination shows short stature, developmental delay, round face, and shortened digits in both hands. Laboratory studies show a low serum calcium concentration and elevated serum phosphate concentration. Genetic studies show mutation in the PTH receptors. She is diagnosed with pseudohypoparathyroidism. Which of the following additional laboratory findings is most likely in this patient?

- A. Decreased urinary phosphate
- B. Increased serum concentration of active vitamin D
- C. Increased serum PTH concentration
- D. Increased serum calcitonin
- E. Decreased serum albumin

7. 46-year-old woman comes to the physician because of a 7-month history of constipation, abdominal discomfort, and fatigue. She was treated for kidney stones 3 months ago. Her vitals are within normal limits. Physical examination shows no other abnormalities. Results of laboratory studies are shown:

Erythrocyte count: 3 million/mm³ (N: 4.5-6)

Serum potassium: 4.5 mEq/L (N: 3.5-5.5)

Serum sodium: 138 mEq/L (135-145)

Serum calcium: 12.9 mg/dL (N: 8.5-10.5)

Fasting blood glucose: 84 mg/dL (N: 70-90)

Serum PTH: Elevated.

Ultrasonography of neck shows well-defined nodular mass in the region of the left parathyroid glands. She is diagnosed with primary hyperparathyroidism due to parathyroid tumor. Which of the following additional laboratory findings is most likely in this patient?

- A. Elevated serum phosphate
- B. Elevated urinary phosphate
- C. Decreased urinary phosphate
- D. Decreased serum bicarbonate
- E. Increased urinary glucose

8. A 10-year-old boy is brought to the physician by his mother because of a 10-month history of bone pains and muscle cramps. Diagnostic studies indicate that he is suffering from vitamin D deficiency. Which of the following sets of alterations is most likely in this patient?

Option	Free serum calcium level	Serum PTH	1-alpha Hydroxylase activity in kidney
A	Low	Low	Low
B	Low	Low	High
C	Low	High	High

D	High	Low	Low
E	High	High	High

9. A 41-year-old woman comes to the physician because of a 3-day history of tingling and numbness in her extremities and recurrent muscle cramps. She had total thyroidectomy for thyroid cancer 1 week ago. Her vital signs are within normal limits. Inflating blood pressure cuff to systolic pressure and keeping for a few seconds produces painful muscle spasm in the hand. Laboratory studies show low serum calcium, high serum phosphate and low urinary phosphate excretion. Which of the following is the most likely diagnosis in this patient?

- A. Primary hyperparathyroidism
- B. Vitamin D intoxication
- C. Vitamin D deficiency
- D. Hypoparathyroidism
- E. Secondary hyperparathyroidism

10. A 67-year-old man comes to the physician because of a 3-week history of difficulty swallowing and a lump in the front of the neck. His brother had neck surgery because of a parathyroid tumor 1 week ago. His vital signs are within normal limits. Physical examination shows a hard mass arising from the upper left lobe of the thyroid on the left side and left-sided cervical lymphadenopathy. Laboratory studies show serum calcium and phosphate concentrations within reference ranges. Examination of the lymph node biopsy shows medullary thyroid cancer. Which of the following tumor markers is the most likely elevated in this patient?

- A. PTH-related peptide
- B. PTH
- C. 1,25-(OH)₂D₃
- D. Calcitonin
- E. Calcitonin gene-related peptide

ANSWER KEY AND EXPLANATION

1. A 61-year-old man comes to the physician because of a 6-week history of bone pain, dyspepsia, and a 3-kg (6.6-lb) weight loss. His vital signs are within normal limits. Physical examination shows dehydration and pallor. Laboratory studies show elevated serum calcium concentration, elevated urinary calcium concentration, and elevated serum PTH concentration. Radiography shows lytic lesions in the long bones and

pelvis. Neck imaging studies show nodular lesion posterior to the right lobe of the thyroid gland. Histological examination of a biopsy specimen from this nodule shows increased mass of the chief cells. Hypercalcemia of this patient is best explained by the action of PTH through PTH receptors located in which of the following organs?

- A. Small intestine
- B. Bones
- C. Skeletal muscles
- D. Liver
- E. Parathyroid gland

Answer: B. Bones. PTH receptors are located only in 2 organs, namely, bones and kidneys. PTH action on the intestine is indirect, through the active form of vitamin D.

2. A 49-year-old man comes to the physician for a follow-up examination. He has an 8-month history of end stage kidney disease who is waiting for a renal transplant. He is having hemodialysis twice a week. He complains of recurrent muscle cramps, bone pains, and fatigue. Laboratory studies show a serum creatinine of 6.1 mg/dL (N: 0.6-1.2), blood urea nitrogen of 76 mg/dL (15-25) and serum calcium of 6.2 mg/dL (N: 8.5-10.5). His glomerular filtration rate is 5 mL/min and has a significant loss of renal function. Which of the following best explains the altered serum calcium in this patient?

- A. Reduced synthesis of cholecalciferol in the skin
- B. Low serum PTH
- C. Low vitamin D intake
- D. Decreased serum albumin
- E. Reduced 1-alpha hydroxylase activity
- F. Reduced 25-hydroxylase activity

Answer: E. Reduced 1-alpha hydroxylase activity. This patient has significant renal damage which resulted in reduced activity of the enzyme, 1-alpha hydroxylase. This enzyme is essential for conversion of inactive vitamin D into active form (calcitriol). Active form of vitamin D acts directly on the gut and increases calcium absorption. Deficiency of this enzyme activity reduces calcium absorption from the gut which leads to hypocalcemia. The other major reason for hypocalcemia in renal failure is the elevated serum phosphate concentration (this is not listed in the options).

3. A 21-year-old man comes to the physician for a routine examination. He says that his elder brother is diagnosed with kidney stones because of a high serum calcium concentration. His vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show a serum calcium of 10.8 mg/dL (Normal: 8.5-10.5) and an elevated serum PTH concentration. A 24-hour urinary calcium excretion is lower than normal. He is diagnosed with a defect in calcium sensing receptors. These receptors are present on which of the following cells?

- A. Oxyphil cells of the parathyroid gland
- B. Chief cells of the parathyroid gland
- C. Parafollicular cells of the thyroid gland
- D. Osteoblasts of bone
- E. Osteoclasts of bone
- F. Small intestinal epithelial cells

Answer: B. Chief cells of the parathyroid gland. Calcium sensing receptors are the transmembrane proteins located on the surface of the chief cells. Serum calcium concentrations are sensed by these receptors and play a vital role in PTH secretion by the parathyroid gland. A low serum calcium sensed by these receptors result in increased secretion of PTH and a high serum calcium suppresses the PTH secretion.

4. A 31-year-old woman comes to the physician because of a 4-week history of recurrent episodes of muscle cramps and sensory disturbance in her hands and legs. She has a 2-year history of malabsorption. She recently moved from tropical to temperate region where she is least exposed to sunlight. Physical examination shows no abnormalities. Laboratory studies show a serum calcium concentration of 8 mg/dL (Normal: 8.5-10.5) and a low serum concentration of vitamin D. She develops muscle cramps when physician asks her to perform hyperventilation for a few minutes. Which of the following mechanisms best explains this patient's presentation following hyperventilation?

- A. Increased PaO₂ reduces ionized serum calcium
- B. Decreased PaO₂ increases ionized serum calcium
- C. Increased PaCO₂ reduces ionized serum calcium
- D. Decreased PaCO₂ reduces ionized serum calcium
- E. Decreased PaCO₂ increases ionized serum calcium

Answer: D. Decreased PaCO₂ reduces ionized serum calcium. This patient has slightly low serum calcium. If this concentration goes further below, she develops symptoms of hypocalcemia such as muscle cramps. This happens when she hyperventilates. Hyperventilation induces alkalosis (because of loss of CO₂) which makes more ionized calcium to bind with the serum albumin. This makes the serum calcium decrease further resulting in muscle cramps.

5. A 26-year-old woman comes to the physician because of an 8-month history of recurrent episodes of muscle cramps, non-bloody diarrhea, and generalized bone pains. She says that diarrhea is accompanied with right lower quadrant pain. Physical examination shows tenderness in the long bones of her upper and lower extremities. Laboratory studies show a serum calcium concentration of 6.8 mg/dL (Normal: 8.5-10.5) and a low vitamin D concentration. Histological examination of a small-bowel biopsy

specimen shows reduced absorptive surface in the small intestinal epithelium. She is diagnosed with Osteomalacia due to reduced absorption of vitamin D in the gastrointestinal tract. Which of the following sets of additional laboratory alterations is most likely in this patient?

Option	Serum PTH	Serum Phosphate	Serum Glucose
A	↑	↑	↑
B	↓	↓	↓
C	↑	↓	↑
D	↓	↑	Normal
E	↑	↓	Normal

Answer: Option E. Osteomalacia is a condition of deficiency of vitamin D in adults. This can occur either because of reduced dietary intake or reduced absorption in GIT. Reduced vitamin D produces reduced active form of vitamin D that is required for the absorption of both calcium and phosphate in the GIT. This leads to reduction in serum concentrations of both calcium and phosphate. Low serum calcium stimulates parathyroid gland to secrete PTH that results in elevated serum PTH. This is called secondary hyperparathyroidism.

6. An 8-year-old girl is brought to the physician by her mother because of a 6-month history of recurrent episodes of muscle cramps. She has a family history of similar symptoms associated with skeletal deformities. Physical examination shows short stature, developmental delay, round face, and shortened digits in both hands. Laboratory studies show a low serum calcium concentration and elevated serum phosphate concentration. Genetic studies show mutation in the PTH receptors. She is diagnosed with pseudohypoparathyroidism. Which of the following additional laboratory findings is most likely in this patient?

- A. Decreased urinary phosphate
- B. Increased serum concentration of active vitamin D
- C. Increased serum PTH concentration
- D. Increased serum calcitonin
- E. Decreased serum albumin

Answer: C. Increased serum PTH concentration. This patient has defective PTH receptors that do not respond to PTH in bones and kidneys. This results in low serum calcium and high serum phosphate. Low serum calcium is sensed by the parathyroid gland and secretes large amount of PTH which increases serum PTH concentration.

7. 46-year-old woman comes to the physician because of a 7-month history of constipation, abdominal discomfort, and fatigue. She was treated for kidney stones 3

months ago. Her vitals are within normal limits. Physical examination shows no other abnormalities. Results of laboratory studies are shown:

Erythrocyte count: 3 million/mm³ (N: 4.5-6)

Serum potassium: 4.5 mEq/L (N: 3.5-5.5)

Serum sodium: 138 mEq/L (135-145)

Serum calcium: 12.9 mg/dL (N: 8.5-10.5)

Fasting blood glucose: 84 mg/dL (N: 70-90)

Serum PTH: Elevated.

Ultrasonography of neck shows well-defined nodular mass in the region of the left parathyroid glands. She is diagnosed with primary hyperparathyroidism due to parathyroid tumor. Which of the following additional laboratory findings is most likely in this patient?

- A. Elevated serum phosphate
- B. Elevated urinary phosphate
- C. Decreased urinary phosphate
- D. Decreased serum bicarbonate
- E. Increased urinary glucose

Answer: B. Elevated urinary phosphate. This patient has excess PTH secretion. PTH inhibits reabsorption of phosphate in the PCT of the kidney. This results in increased phosphate loss in urine and reduced serum phosphate concentration.

8. A 10-year-old boy is brought to the physician by his mother because of a 10-month history of bone pains and muscle cramps. Diagnostic studies indicate that he is suffering from vitamin D deficiency. Which of the following sets of alterations is most likely in this patient?

Option	Free serum calcium concentration	Serum PTH	1-alpha Hydroxylase activity in kidney
A	Low	Low	Low
B	Low	Low	High
C	Low	High	High
D	High	Low	Low
E	High	High	High

Answer Key: C.

Vitamin D is required for absorption of calcium from gut. Low vitamin D levels would decrease plasma free calcium, and this in turn increases serum PTH (secondary hyperparathyroidism). Elevated PTH increases 1-alpha hydroxylase activity in the kidney.

9. A 41-year-old woman comes to the physician because of a 3-day history of tingling and numbness in her extremities and recurrent muscle cramps. She had total

thyroidectomy for thyroid cancer 1 week ago. Her vital signs are within normal limits. Inflating blood pressure cuff to systolic pressure and keeping for a few seconds produces painful muscle spasm in the hand. Laboratory studies show low serum calcium, high serum phosphate and low urinary phosphate excretion. Which of the following is the most likely diagnosis in this patient?

- A. Primary hyperparathyroidism
- B. Vitamin D intoxication
- C. Vitamin D deficiency
- D. Hypoparathyroidism
- E. Secondary hyperparathyroidism

Answer key: D.

Low blood Ca^{2+} and high blood phosphate are consistent with low serum PTH (hypoparathyroidism). Lack of parathyroid hormone (PTH) decreases bone resorption, decreases renal reabsorption of Ca^{2+} , and favors renal reabsorption of phosphate (causing low urinary phosphate). Vitamin D intoxication would cause hypercalcemia, not hypocalcemia. Vitamin D deficiency would cause both hypocalcemia and hypophosphatemia because active form of vitamin D is required for GI absorption of both calcium and phosphate.

10. A 67-year-old man comes to the physician because of a 3-week history of difficulty swallowing and a lump in the front of the neck. His brother had neck surgery because of a parathyroid tumor 1 week ago. His vital signs are within normal limits. Physical examination shows a hard mass arising from the upper left lobe of the thyroid on the left side and left-sided cervical lymphadenopathy. Laboratory studies show serum calcium and phosphate concentrations within reference ranges. Examination of the lymph node biopsy shows medullary thyroid cancer. Which of the following tumor markers is the most likely elevated in this patient?

- A. PTH-related peptide
- B. PTH
- C. $1,25\text{-(OH)}_2\text{D}_3$
- D. Calcitonin
- E. Calcitonin gene-related peptide

Answer key: D.

Calcitonin is secreted from the parafollicular cells (C cells) of the thyroid gland. Physiologically, calcitonin lowers serum calcium by inhibiting bone resorption and increasing renal calcium excretion. However, these actions are very weak and do not contribute to serum calcium homeostasis. Therefore, in medullary thyroid cancer, even though serum calcitonin levels are high, serum calcium and phosphate levels usually remain normal. Calcitonin is a tumor marker in the diagnosis of medullary thyroid cancer.

